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Aim : Aim: Simulation of TDMA-based MAC Protocol

Simulate TDMA slot allocation and compare energy efficiency with CSMA.

```
BEGIN {
    sent = 0
    received = 0
    sent_bytes = 0
    received_bytes = 0
    total_delay = 0
}

{
    event = $1
    time = $2
    from = $3
    to = $4
    packet_type = $7
    seq_num = $6
    packet_size = $8

    # Packet sent
    if (event == "+" && packet_type == "cbr") {
        sent++
        sent_bytes += packet_size

        send_time[seq_num] = time
    }

    # Packet received
    if (event == "r" && packet_type == "cbr") {
        received++
        received_bytes += packet_size

        if (seq_num in send_time) {
            delay = time - send_time[seq_num]
            total_delay += delay
        }
    }
}

END {

    # Packet Delivery Ratio
    if (sent > 0) {
```

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```
pdr = (received / sent) * 100
}
else {
    pdr = 0
}

# Throughput in bits per second
simulation_time = 8.0 - 1.0

throughput = (received_bytes * 8) / simulation_time

# Average end-to-end delay
if (received > 0) {
    avg_delay = total_delay / received
}
else {
    avg_delay = 0
}

print "=====
print "    MANET PERFORMANCE RESULTS"
print "=====

printf "Packets Sent      : %d\n", sent
printf "Packets Received   : %d\n", received

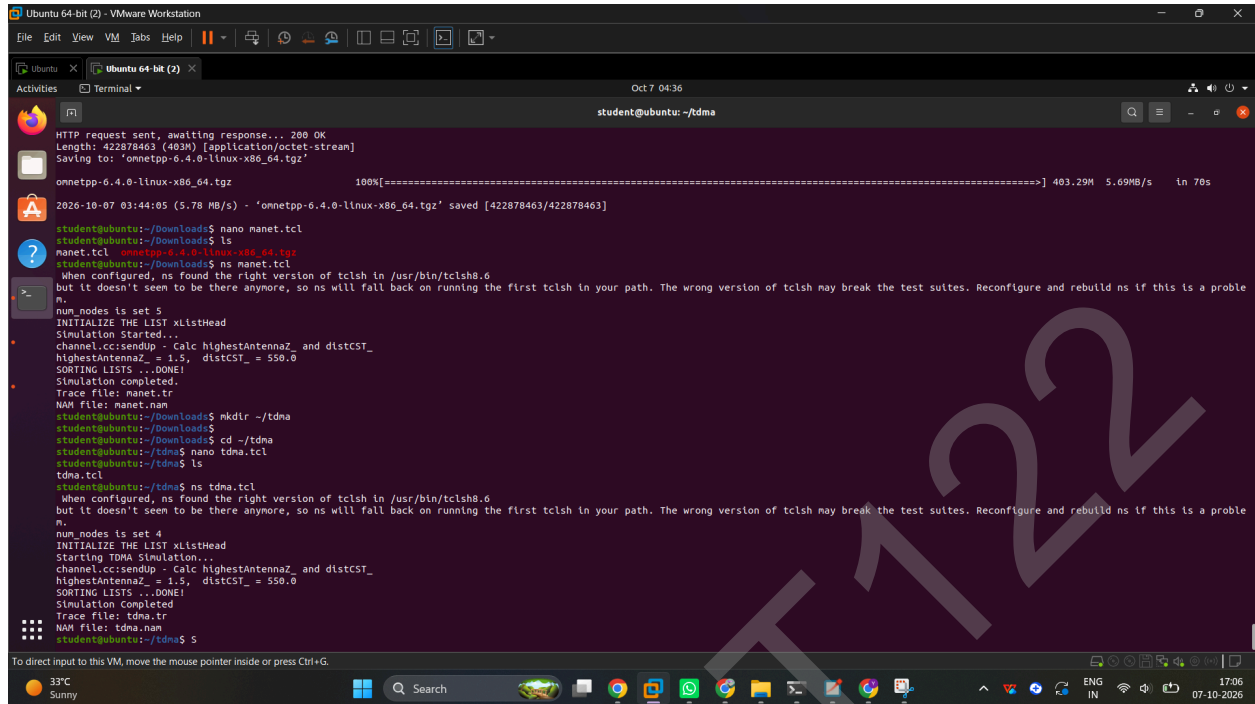
printf "Packet Delivery Ratio : %.2f %%\n", pdr

printf "Throughput          : %.2f bits/sec\n", throughput

printf "Average End-to-End Delay : %.6f sec\n", avg_delay

print "=====
}
```

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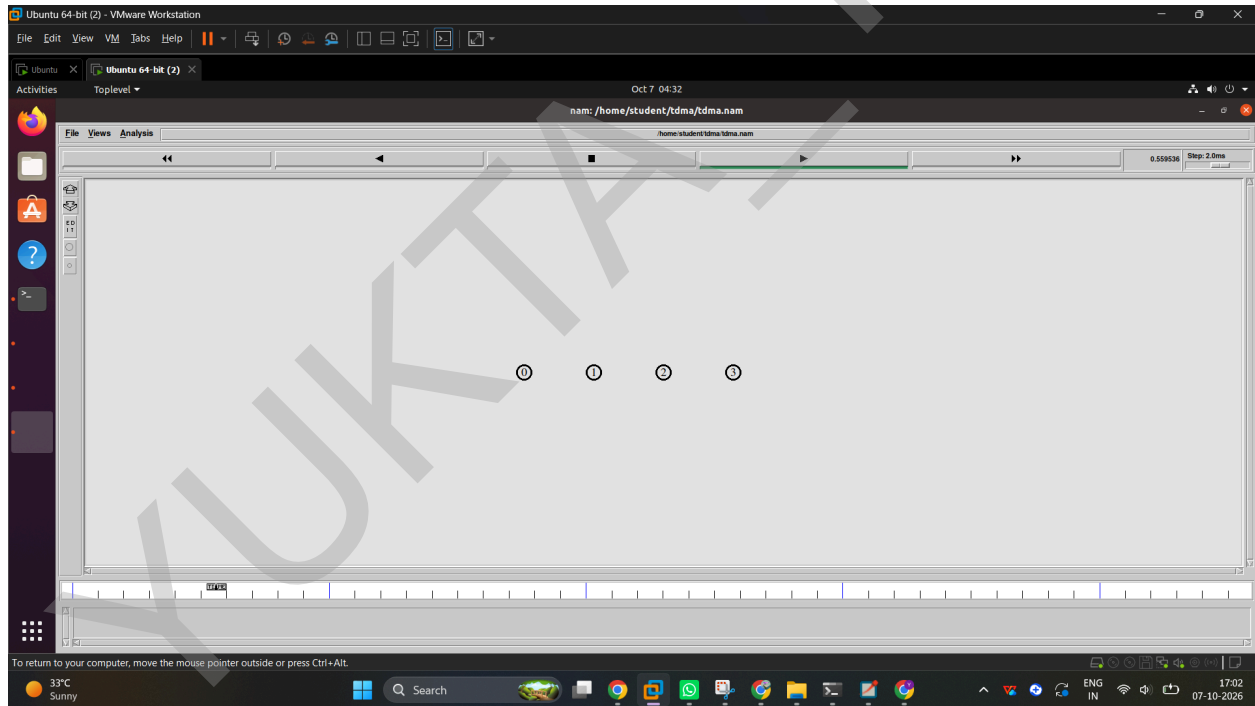


```
student@ubuntu: ~/tdma
HTTP request sent, awaiting response... 200 OK
Length: 422878463 (403M) [application/octet-stream]
Saving to: 'onnetpp-6.4.0-linux-x86_64.tgz'

onnetpp-6.4.0-linux-x86_64.tgz      100%[=====] 403.29M  5.69MB/s   in 70s

2026-10-07 03:44:05 (5.78 MB/s) - 'onnetpp-6.4.0-linux-x86_64.tgz' saved [422878463/422878463]

student@ubuntu: ~/Downloads$ nano manet.tcl
student@ubuntu: ~/Downloads$ ls
manet.tcl  onnetpp-6.4.0-linux-x86_64.tgz
student@ubuntu: ~/Downloads$ ns manet.tcl
When configured, ns found the right version of tclsh in /usr/bin/tclsh8.6
but it doesn't seem to be there anymore, so ns will fall back on running the first tclsh in your path. The wrong version of tclsh may break the test suites. Reconfigure and rebuild ns if this is a problem.
num_nodes is set 5
INITIALIZE THE LIST xListHead
Simulation Started...
channel.cc:sendup - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS --DONE!
Simulation completed.
Trace file: manet.tr
NAH file: manet.nam
student@ubuntu: ~/Downloads$ mkdir ~/tdma
student@ubuntu: ~/Downloads$ cd ~/tdma
student@ubuntu: ~/tdma$ nano tdma.tcl
student@ubuntu: ~/tdma$ ls
tdma.tcl
student@ubuntu: ~/tdma$ ns tdma.tcl
When configured, ns found the right version of tclsh in /usr/bin/tclsh8.6
but it doesn't seem to be there anymore, so ns will fall back on running the first tclsh in your path. The wrong version of tclsh may break the test suites. Reconfigure and rebuild ns if this is a problem.
num_nodes is set 4
INITIALIZE THE LIST xListHead
Starting TDMA Simulation...
channel.cc:sendup - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS --DONE!
Simulation Completed
Trace file: tdma.tr
NAH file: tdma.nam
student@ubuntu: ~/tdma$ S
```



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